Hardware Project No. 5
Not-Quite Pulse Oximeter

ECE 444: Bioinstrumentation and Sensing
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1. Introduction

The goal of this project was to build a simple optical pulse detector using a red LED on one side of a fingertip and a photodetector on the other side. The circuit is not a full pulse oximeter because it does not compare red and infrared absorption or compute blood oxygen saturation. Instead, it only attempts to observe the small pulsatile change in transmitted light caused by changing blood volume in the finger.

2. Circuit Design

The final design used a red LED as the light source and a phototransistor as the detector. The phototransistor was chosen after initial testing because it gave a stronger and easier-to-use signal than the photodiode and photocell trials. The detector output contains a large DC component from the steady light level plus a much smaller AC pulse component. Therefore, an RC coupling network and LF411 op-amp amplifier were used to make the heartbeat waveform easier to see on the oscilloscope.

The reference circuit used for the design is shown in Figure 1. This circuit was adapted from the Union College BME/ECE 386 Lab 4 optical heart-rate monitor handout. I kept the same general structure: LED and phototransistor across the fingertip, an RC high-pass coupling stage, and a non-inverting op-amp amplifier with a feedback capacitor for low-pass filtering.

The main practical change from the reference circuit was that the op-amp was built with an LF411 available in the lab and powered from the available dual supply rails rather than the exact +/-5 V rails shown in the reference image. The LED and phototransistor were also implemented with the actual parts found in the lab bins and adjusted experimentally so the detector responded to light and dark conditions before trying the fingertip measurement.

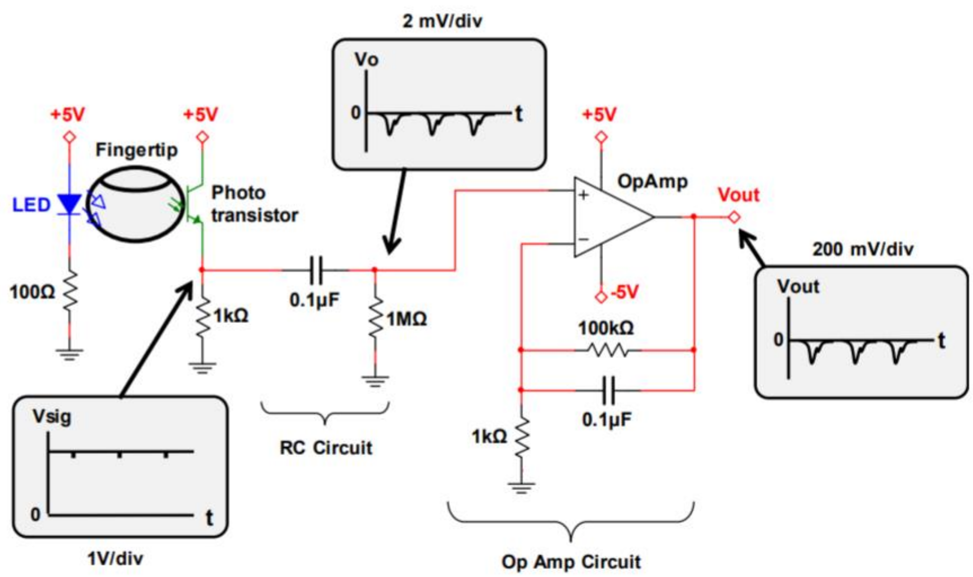

Figure 1. Reference optical heart-rate circuit used as the basis for the breadboard implementation, adapted from the Union College BME/ECE 386 Lab 4 optical heart-rate monitor handout. The important parts copied were the phototransistor detector, the 0.1 uF and 1 MOhm RC coupling network, and the non-inverting op-amp amplifier/filter using a 100 kOhm feedback resistor, 0.1 uF feedback capacitor, and 1 kOhm resistor to ground.

3. Experimental Procedure

First, the LED and phototransistor were tested without the op-amp to confirm that the detector voltage changed when the phototransistor was exposed to light or covered. The LED and phototransistor were then placed on opposite sides of the fingertip. The signal was AC-coupled through the RC network into the LF411 amplifier.

The oscilloscope was connected to the output of the amplifier to look for a slow periodic waveform corresponding to the heartbeat.

The physical breadboard layout is shown in Figure 2. Because the signal is optical, the exact alignment of the LED, fingertip, and phototransistor mattered. The detector was much more useful once the parts were aimed at each other and the finger was placed directly between them.

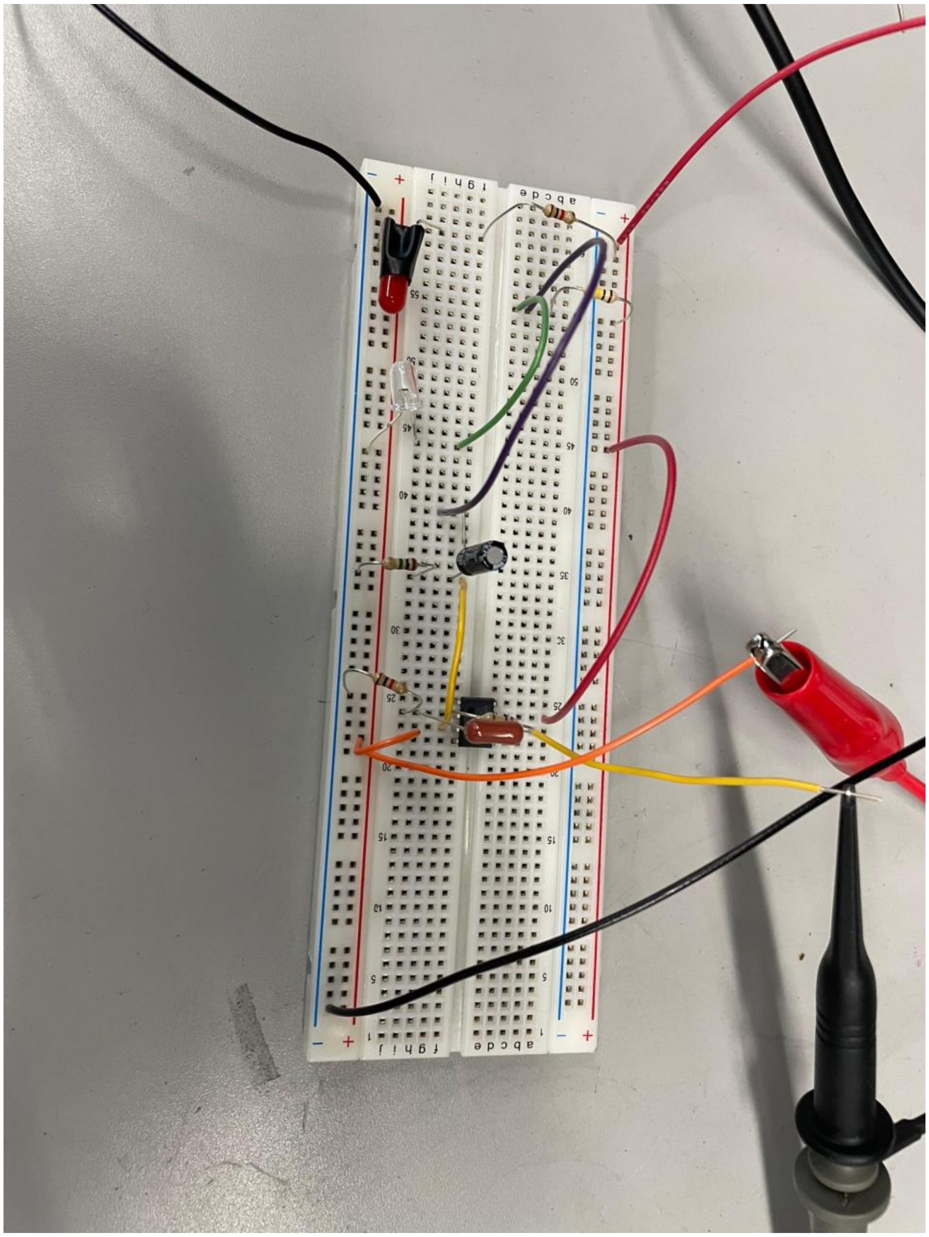

Figure 2. Breadboard implementation of the LED, phototransistor detector, RC coupling network, and LF411 amplifier circuit.

4. Results

With the fingertip placed between the red LED and the phototransistor, the oscilloscope showed a repeating waveform at the output of the amplifier. The waveform was not perfectly clean, but it was periodic and consistent with a pulse signal. The oscilloscope screenshot in Figure 3 shows the measured output after amplification. The displayed peak-to-peak measurement was approximately 2.3 V on channel 2.

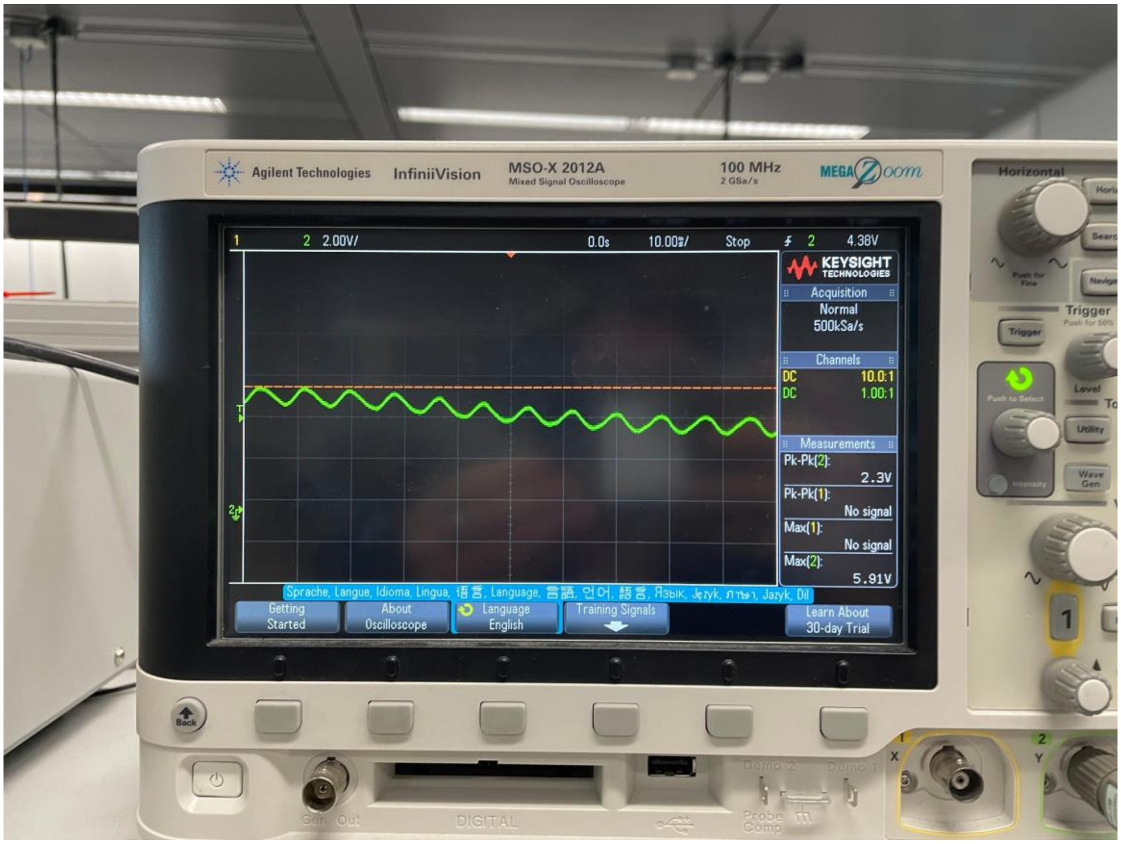

Figure 3. Oscilloscope output with fingertip placed between the red LED and phototransistor. The observed waveform is periodic and was used as the measured pulse signal.

5. Discussion and Conclusion

The completed circuit demonstrated the basic principle of a not-quite pulse oximeter. The red LED transmitted light through the fingertip, and the phototransistor converted the received light variation into an electrical signal. The RC network removed much of the large DC component, and the LF411 non-inverting amplifier increased the small pulse-related variation enough to observe it on the oscilloscope.

The main limitation was that the optical setup was very sensitive to alignment, finger pressure, and ambient light. If the detector received too much direct light, the output saturated low; if the finger blocked too much light, the detector output moved close to the dark condition. The amplifier and filtering stage were therefore necessary to make the small heartbeat-related ripple visible. Overall, the circuit successfully produced a measurable periodic waveform when the fingertip was placed between the LED and phototransistor, satisfying the main goal of the project.

References

Kremenic, A. ECE 444 Bioinstrumentation and Sensing, Hardware Project No. 5: Not-Quite Pulse Oximeter. The Cooper Union, 2026.

Buma, T. BME/ECE 386 Lab 4: Optical Heart Rate Monitor. Union College, 2020. Available at: https://minerva.union.edu/bumat/Teaching/BME386/Labs/Lab4_BME386_2020.pdf